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Human exit choice in crowded built environments: Investigating underlying behavioural differences between normal egress and emergency evacuations

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ABSTRACT

Egress behaviour of pedestrians in crowded complex confined spaces is investigated in this study. Despite recent methodological progress in the development of simulation tools for predicting crowd egress and evacuation, little is known based on empirical data about the underlying rules that govern exit wayfinding of pedestrians in multi-exit places. Particularly, fundamental differences between behavioural features of emergency and non-emergency egress have not been fully explored by previous studies. Stated-choice data was collected in face-to-face interviews with passengers as they exited a major railway station in Melbourne. Participants were asked what exit decision they would have made given certain hypothetical scenarios at that same station. Econometric models (error-component mixed logit) were developed to quantify the way passengers evaluate and prioritise various contributing factors while accommodating the potential decision heterogeneity. These factors include distance, crowding, visibility of exits, proximity of the exits to their destination, impact of other passengers' decisions; and spatial distribution of exits. Key findings of our modelling suggest that (1) for nonemergency egress, proximity of the exit points to passenger's destination is a dominant factor although not the sole determinant. (2) In an emergency, passengers place a much higher priority on avoiding crowded exits compared to non-emergency situations. (3) Directional flows of pedestrians do not significantly impact on decisions made in a normal egress. (4) In an emergency evacuation, directional flows are considered as a negative utility factor by majority of individuals, although the perception of this factor is highly heterogeneous and also depends on the visibility of the exits targeted by the flow. The proposed static models can be incorporated with a broad range of crowd simulation methods as alternatives for heuristic modeller-defined exit choice rules. They accommodate the dissimilarities of egress behaviour between emergencies and non-emergencies, thereby enabling planners to conduct virtual simulations under either scenario.

1. Introduction

Population growth as well as the increasing occurrence of large public events, such as sport events and entertainment programs which gathers massive numbers of people, have subjected public transport facilities to unprecedented levels of demand and congestion. Moreover, the possibility of the need for emergency evacuations of those facilities as a result of terrorist activities, fires or accidental calamities have made authorities concerned about providing the safest and most efficient architectural designs or management solutions. The tragic events within the last several decades during which crowd mismanagement has caused disasters and casualties highlight the importance of this issue [23]. Crowd panic and stampedes can have disastrous consequences, as the 2010 Germany Love Parade Disaster sadly demonstrated, 21 people were killed and another 510 seriously injured. In retrospect, many such real crowd disasters could have been avoided with better crowd management. In line with the increasing attention to the general topic of crowd management, the literature has also exhibited a particular interest in investigating this issue in conjunction with the safety of major public transport stations [12,15,38,9] as facilities that serve large numbers of pedestrians on a regular basis. Fridolf et al. [12] provide a review on previously-reported fire accidents in underground transport systems in addition to evaluating four different human behaviour theories that can be used in fire safety design of railway stations.

Clearly, prevention of such incidents entails development of

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robust modelling tools and methodologies to predict crowd behaviour during planning and design stages of public transport stations or public buildings. The ability to conduct virtual simulations on the operation of such facilities under different possible circumstances such as extreme levels of demand or emergency situations will enable authorities to plan and prepare for worst-case scenarios and estimate evacuation (or discharge) time, thereby improving safety assessment of crowded enclosed spaces. Accordingly, crowd simulation now plays a central role in support of planning and decision making for public safety. Although this has become a rather well-established field of research and has led to the development of several commercial simulation tools, many important aspects of crowd behaviour are not well understood, particularly in extreme emergency and panic scenarios [32]. Panics are unique because individual cognitive abilities may dramatically reduce and behavioural features of people such as movement speeds are taken to extremes. For all those reasons, it is believed that the accuracy and reliability of the existing crowd simulation programs for practical planning purposes are still limited [44]. It is also believed that the lack of concrete knowledge in this area of research chiefly stems from the lack of reliable empirical data through which the behaviour of individuals can be reflected into the forecast tools [34]. The potential heterogeneity of the behaviour that different individuals may exhibit in a certain egress or evacuation scenario also adds to the complexity of the problem.

Particularly, the insufficiency of the existing empirical evidence in this field has left many unanswered questions with regard to the wayfinding of passengers when exiting or evacuating architecturally-complex built environments such as public transport stations. Do pedestrians consider a single criterion for choosing their exit or there is a combination of factors involved? If the latter is the case, what are those contributing factors and their relative weights on their decision making? Do these potential weights vary significantly from person to person? Are there fundamental differences between such factors and their combined effect between daily exit choices of passengers and those of evacuation situations? If so, how one can quantify those differences and capture them in the forecast process? These constitute some of the main questions this research attempts to address.

In this work, a random-utility analysis is conducted, as one of the theoretically rigorous and well-accepted methodologies of modelling human decision. Models are developed based upon stated-choice data sets collected in a major railway station in Melbourne city. Of particular interest is identifying the fundamental factors influencing passengers' route choice behaviour in complex (multi-exit) geometries. Our aim is to understand and quantify the difference between the ways pedestrian passengers evaluate contributing factors in complicated exit-choice situations. By complex exit choice scenarios we refer to the situations when there is a trade-off between different competing factors such as choosing less-congested exit points versus closer exit points. We also make a distinction between exit choices of passengers in normal (nonemergency) and emergency situations. In addition to quantifying passengers' trade-off between congestion and distance, we also attempt to understand if passengers tend to follow other peoples' route decisions and under what circumstances they may choose to head to an invisible exit point whose level of crowding is uncertain to them. We also examine the impact of proximity of exit points to destinations and spatial distribution of exits on passengers' decisions. The potential heterogeneity of the passenger behaviour is also accommodated through application of random-coefficient discrete choice models. The remainder of the paper is structured as follows. In Section 2, we make a selective review on the current practice and research on modelling behaviour of passengers in railway stations with a particular emphasis of the topic of egress wayfinding and exit choice. The limitations of the methods currently practised in the literature are also briefly discussed in this section. The data collection and survey procedure are explained in Section 3. Theoretical background of our modelling, the model estimation results and our behavioural interpretations are discussed in Section 4. Sections 5 and 6 are respectively dedicated to a discussion and summary of the findings.

2. State of the art and limitations of the current practice

Extensive research within recent years has been carried out to enhance the safety strategies of crowded railway and public transport stations by providing better understanding of different aspects of the passenger behaviour [43,5,52]. The problem has been viewed from different perspectives such as simulation of passengers alighting and boarding [57], issues related to passengers' use of stairways versus escalators during level changing [29,7] or investigating particular features of crowd behaviour in complex manoeuvres like turning and merging points [2,51]. As an exploratory study, Gräßle and Kretz [16], dell'Olio et al. [8] and [13] also investigated train evacuation behaviour of passengers (and crews) by conducting mock evacuation experiments under different conditions of exit configurations or conducting stated choice experiments to quantify the behavioural differences associated with different types of incidents. The literature on this topic, however, is dominated by the application of the existing methods or commercial software of crowd emergency evacuation [36,56,58]. Cheng and Yang [6] conducted case studies to evaluate capacities of certain subway stations in China. Jiang et al. [30] also conducted case studies with the aim of tuning of two certain input parameters of a commercial crowd simulation software (maximum upstairs speed and the average minimum width of staircase utilised per person) to reflect particular characteristics of passengers flow in railway stations by taking two subway stations in China as examples.

Although crowd simulation is a well-established field of research , and several commercial software packages are available, many important aspects of crowd behaviour are not well understood [35]. Particularly, the underlying difference between passengers' wayfinding behaviour in extreme emergency (panic) scenarios and normal egress situations has barely been investigated by earlier studies. One major reason for that is the scarcity of dedicated empirical data, both in terms of quality and quantity, which hampers validation of existing models or calibration of input parameters [34,44]. Also, limited attention has been paid to the necessity of application of stochastic behavioural models that can accommodate the diversity as well as the uncertainty of human escape behaviour [45]. A few former studies have partially addressed the problem of lack of explanatory data by providing choice data [10,18,20–22,39], conducting experimental studies in relatively simple geometries [24] or monitoring actual movements of pedestrians in large crowds [16]. However, none of those have addressed the particular problem of wayfinding by railway passengers nor have they made clear distinction between the potential dissimilarities of route-finding behaviour between normal and emergency egress situations.

On the particular topic of crowd emergency exit choice, four general major approaches have commonly been practised in the literature thus far. These include the game theory approach [11,37,40], discrete-choice methods [1,10,18,39], network-based models [3] and implementation of implicit rules in transition probabilities of cellular-automaton-based methods [27]. Kneidl et al. [31] also reported on the development of an approach that integrates a network-based method for exit choice (i.e. large-scale navigation also known as "tactical decisions") with a cellular-automata simulation framework. Many of those methods, however, do not make use of any empirical data and are purely based on assumptions. Among the studies which have provided empirical evidence for their modelling or statistical analysis, some have provided insights through conducting evacuation experiments under control conditions [24,42]. Another common practice has been application of different forms of virtual exit choice experiments [4,42] notably in the specific form of stated choice experiments [10,39].

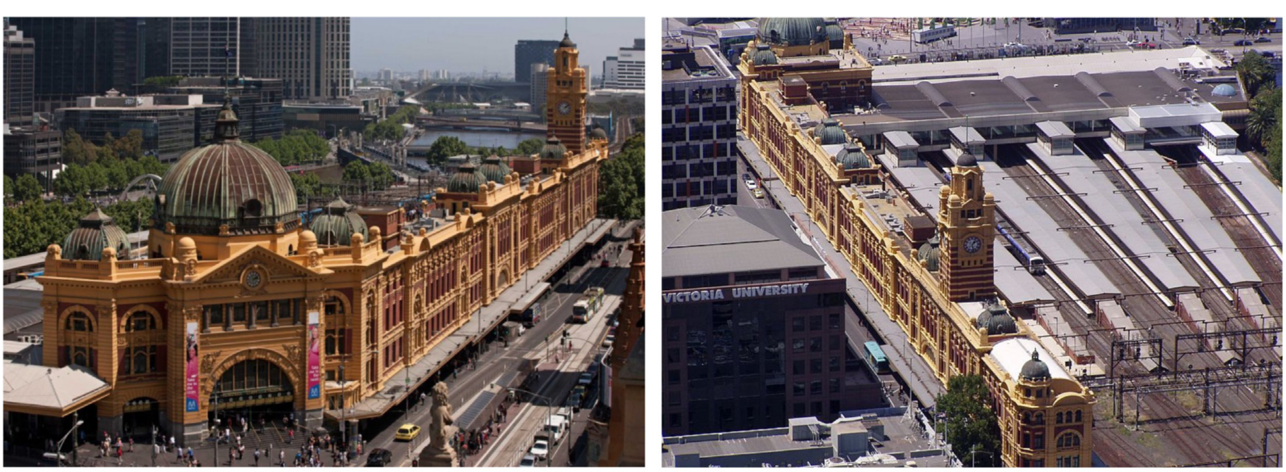
Despite those efforts, however, there are some important aspects of the exit choice problem that have often been taken for granted in the previous studies. Among those, is the effect of factors such as spatial distribution of exits, exit visibility, the impact of other pedestrians' decisions (known as "herding" behaviour) and particularly the interactive effect of herding and exit visibility. The binary choice scenarios designed in some previous studies [10,39] do not essentially allow investigating the effect of spatial distribution of the exits to be captured by the model. Also the concept of visibility, to our knowledge, has not been examined in conjunction with the way it interacts with the tendency of pedestrians to follow others' decisions. There have been studies to investigate the effect of low visibility conditions on evacuation behaviour [14,17,28,41] by conducting experiments while having participants wear eyepatches or filling the experiments area with smoke. However, the fact that even in non-smoke-filled conditions of evacuation the architectural design of the egress environment makes some exits invisible to the decision maker has not been fully appreciated in the former works. Therefore, the question will arise as to the conditions which make an individual evacuee prefer an invisible exit over an exit whose conditions (i.e. the level of crowding etc.) are certain to the individual. Also whether this factor interacts with the way evacuees make or change their decisions based on others' decisions is yet to be explored. To address the effect of herding, Korhonen and Heliövaara [33] classified evacuees into the ones who follow other pedestrians and the ones who do not. Whether such classification exists in the population of evacuees or not of course requires investigation based on experimental data [19]. Furthermore, it is arguable if the herding effect per se could be considered as a sole determinant for one individual to make their decision as it seems to be more reasonable to assume that there is a combination of various factors that plays the key role in exit choice decisions of individuals. This study is intended to explore the existence of such combination and the relative weights (importance) of different contributing factors. How such combination is different between the exit decisions made in a normal (everyday) egress situation and those made in an emergency also constitutes another key element of this study.

3. Data and survey

Stated choice experiments were carried out by face-to-face interviewing passengers in Flinders Station in Melbourne (Fig. 1) as they exited the station. Flinders Street Station, Australia's busiest rail station accommodates an average of 100,000 passengers on a daily basis but with significantly higher crowds during special events. Also, in terms of the architectural design, it offers multiple exit points to passengers. For the survey design purpose, this provides a good possibility for efficient hypothetical choice scenarios to be designed with reference to the geometry of the station. Hypothetical choice scenarios were designed over a simplified representation of the station's map illustrated in Fig. 2(a).

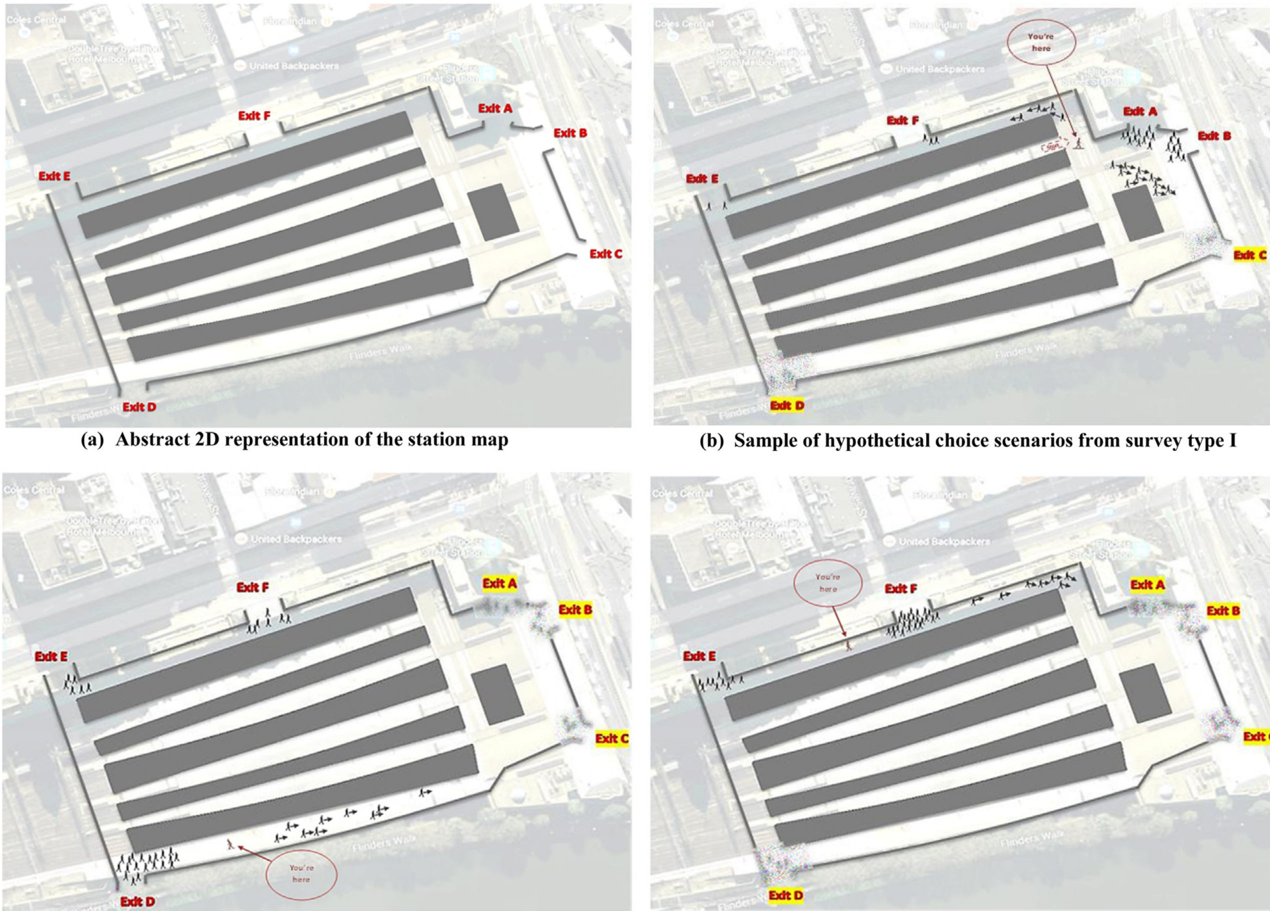
The data collection methodology practiced in this study is essentially similar to the approach taken by some former studies in the field of pedestrian evacuation [10,39]. The main difference that the current work intends to make in terms of the data provision method is that, rather than introducing people to purely hypothetical choice scenarios, we design the choice experiments with reference to the actual choices that respondents recently made and of which they have a clear memory. This is in line with a new trend in econometrics literature in relation with application of stated-choice methods to enhance realism of hypothetical choice experiments [46,53]. Pivoting the hypothetical scenarios with a direct reference to the person's recent real choice is believed to make participants relate more realistically to the experiments, pay more attention to the survey scenarios and as a result, provide more reliable data. Research has also shown that this method of stated choice data collection, often known as SP-off-RP (stated preference off revealed preferences) is more likely to lead to more reliable data and more efficient model estimation results [55]. Further, the fact that the data is collected by face-to-face interviews brings about another potential benefit. Reasonably, compared to more prevalent and time-efficient approaches such as online surveys, interviews are believed to be more effective in terms of preventing distraction and inattention by respondents which has been known as a major issue in relation with data collected in the form of stated choices [25].

The station offers six exit points to passengers (see Fig. 2(a)). We have labelled them as Exit A to Exit F. Three series (types) of choice scenarios were designed and each passenger interviewee was introduced to the specifically-designed scenarios



(a) (b)

Fig. 1. Flinders Street Station in Melbourne City.



(a) Abstract 2D representation of the station map
(b) Sample of hypothetical choice scenarios from survey type I
(c) Sample of hypothetical choice scenarios from survey type II
(d) Sample of hypothetical choice scenarios from survey type III

Fig. 2. Samples of Hypothetical Choice Scenarios. (a) Abstract 2D representation of the station map. (b) Sample of hypothetical choice scenarios from survey type I. (c) Sample of hypothetical choice scenarios from survey type II. (d) Sample of hypothetical choice scenarios from survey type III.

corresponding to their latest real exit choice. We refer to those as survey types I, II and III. Passenger interviewees who were observed to have exited the station via exit A, B, or C were introduced to choice scenarios of type I, the ones who have exited the station through exit D were introduced to choice scenarios of type II; and those exited from exits E or F responded to scenarios of type III. Survey type I and III each consist of fourteen hypothetical choice experiments and survey type II consists of ten scenarios. Fig. 2(b)–(d) illustrate one example of each scenario type.

In each choice experiment, the hypothetical position of the passenger interviewee, the level of crowding around each exit point and also the number of other pedestrians moving towards exit points are represented and varied over the scenarios. These scenarios basically investigate the conditions under which passengers may change their recent exit choice to another exit alternative. In the survey design, it is taken into account the exit points that are not visible to the decision maker from their current hypothetical position. The area around each invisible exit point (highlighted in each picture) was blurred out indicating ambiguity in the level of crowding around them. Participants were told that they can choose any of the six exits, whether visible or invisible, in each scenario. The only difference is that the level of congestion around invisible exits is not given to them. By changing the hypothetical position of decision maker over the experiments, the distance to different exit points as well as the visibility condition of different exits is varied.

Participants were asked to state their choices in each hypothetical scenario for two different conditions. One response was stated assuming that the decision is being made during an everyday (normal) egress situation and in the other; they were asked to assume that the exit is taking place under an emergency. In total, 105 interviews were conducted which divides to 65, 15 and 25 interviews using survey types I, II and III respectively. This provides us with two choice data sets of 1410 observations.

4. Modelling

4.1. Model specification

Utility of each exit point i for decision maker n is formulated as Eq. (1), in which V_{nit} represents the deterministic (representative) part of utility and ε_{nit} signifies the random error component of utility. The subscript t signifies the choice scenario number t in the sequence of the choice scenarios to which each individual participant has responded. The random error terms are assumed to be identically and independently[1] distributed as Extreme Value Type I [18]. They reflect all the factors not being captured by the deterministic part of utilities and accommodate the stochastic nature of decisions. The decisions are modelled based upon the underlying utility maximisation axiom assuming that each individual choose the alternative from which they perceive the highest utility. Therefore, the probability of choosing alternative i is generally given by Eq. (2).

[1] Independency is assumed to exist over alternatives, individuals and choice experiments.

$$U_{nit} = V_{nit} + \varepsilon_{nit} \tag{1}$$

$$P_{nit} = Prob\left(U_{nit} > U_{njt};\forall j \neq i\right) = Prob\left(\varepsilon_{njt} - \varepsilon_{nit} < V_{nit} - V_{njt};\forall j \neq i\right) \tag{2}$$

Eq. (3) specifies the representative part of the utility in our application that includes the explanatory variables that we have applied to our particular problem. The definitions of the explanatory variables are also given as follows.

$$\begin{aligned} V_{nit} = {} & \beta_{1n}\left(DIST_{nit}\right) + \beta_{2n}\left(CROWD_{nit}\right) + \beta_{3n}\left(DIRFLOW1_{nit}\right) \\ & + \beta_{4n}\left(DIRFLOW2_{nit}\right) + \beta_{5n}\left(VIS_{nit}\right) + \beta_{6n}\left(DEST_{nit}\right) \\ & + \beta_{7n}\left(NEST1_{it}\right) + \beta_{8n}\left(NEST2_{it}\right) \end{aligned} \tag{3}$$

$DIST_{nit}$: distance of individual n to exit point i in choice scenario t;

$CROWD_{nit}$: crowding (number of pedestrians) around exit point i in choice experiment t;

$DIRFLOW1_{nit}$: equals the number of pedestrians (directional flow), visible by individual n and headed towards exit point i if exit i is visible to individual n in choice experiment t, otherwise (i.e. if the exit is invisible) it equals 0;

$DIRFLOW2_{nit}$: equals the number of pedestrians (directional flow), visible by individual n and headed towards exit point i if exit i is invisible to individual n in choice experiment t, otherwise (i.e. if the exit is visible) it equals 0;

VIS_{nit}: binary variable which equals 1 if exit point i is visible to individual n in choice experiment t, and otherwise it equals 0;

$DEST_{nit}$: binary variable which equals 1 if exit point i in experiment t leads to individual n's observed destination; otherwise it equals 0;

$NEST1_{it}$: binary variable which equals 1 if exit i in experiment t is one of the exits A, B or C, and equals 0 otherwise;

$NEST2_{it}$: binary variable which equals 1 if exit i in experiment t is one of exits E or F, and equals 0 otherwise.

Also, β's represent the utility coefficients. We employ a random-coefficient multinomial logit model which allows coefficients to vary from individual to individual. This specification allows the modeller to elicit any possible heterogeneity (variation) in the values that different people place on different factors that affect their exit choices. Moreover, it also allows the effect of serial correlation to be reflected into the modelling process. This refers to the fact that in this case we are dealing with panel choice data sets in which each individual provides several choice observations. The presence of random coefficients in this type of model specification accommodates this effect. Formal mathematical proof of this can also be found in other references [50]. Here we assume utility coefficient to be distributed normally over the sample of individuals[2].

The probability that each individual chooses the same sequence of alternatives that they did in the sample, $\boldsymbol{i}=(i_1, i_2,\ldots)$, is given by Eq. (4), in which $\boldsymbol{\beta}$ signifies the vector of utility coefficients and $f(\boldsymbol{\beta})$ shows the probability density function of those coefficients. The integral cannot be solved analytically and the common practice is to evaluate the integral using simulation methods [54]. The likelihood function which yields the estimated parameters is also given in Eq. (5).

$$P_{nit} = \int \prod_t \frac{e^{V_{ni_t t}}}{\sum_j e^{V_{njt}}} f(\boldsymbol{\beta}) d\boldsymbol{\beta} \tag{4}$$

[2] Other alternative types of distribution such as log-normal, triangular, and uniform and also their possible combinations have been examined during the data mining process, however, none of which led to any better statistical fit than the model based on normal distribution assumption.

$$Likelihood\ \ Function = \prod_n P_{nit} = \prod_n \left(\int \prod_t \frac{e^{V_{ni_t t}}}{\sum_j e^{V_{njt}}} f(\boldsymbol{\beta}) d\boldsymbol{\beta} \right) \tag{5}$$

4.2. Model estimation results

Results of the model estimation for both normal egress model and emergency egress model are presented in Table 1. The measures of statistical fit for both models (0.35 and 0.34) are fairly large values for a typical choice model. Statistical significance of the estimated utility coefficients for each variable (indicated by *'s) provides information as to which variable plays a significant role in exit decisions under each scenario (i.e. normal and emergency). Also, the absolute value of the estimates determines the relative significance of each factor between the two models. In the following sections we discuss and interpret the behavioural findings that can be obtained based on the model estimation results by putting into contrast the results from the normal and emergency egress models.

4.3. Behavioural interpretation of the results

4.3.1. Proximity to destination

The results strongly suggest that in normal wayfinding of passengers, proximity to destination is a dominant factor whereas for emergency egress, as expected, this factor does not play a meaningful role in exit decisions. The estimated coefficient for this variable (DEST) in the normal-egress model proves to be a large number which is a clear indication of the key role this factor plays in everyday exit decisions of passengers. Yet, as the normal-egress model suggests, this factor is not the sole determinant in normal exit choices and it is the combination of the exit utility factors (such as crowding and distance etc.), relative importance of which given by the model, that determine the choice. In other words, as suggested by the model, certain combinations of the exit attributes (e.g. certain levels of crowding around exits) might make the decision maker choose an exit that is not the closest to their destination in order to avoid excessive delay.

Inclusion of this variable in the emergency model, however, leads to insignificant estimations for the associated coefficient and also damages the models' goodness of fit. Also, theoretically it does not stand to reason to include this variable for the emergency model as it goes without saying that in such situations the sole purpose (i.e. destination) of pedestrians will be merely leaving the enclosed space. For that reason, we forced the coefficient of DEST to be zero in the emergency model.

4.3.2. Spatial distribution of exits

The model suggests that the spatial distribution of exit points have a more significant impact on passengers' emergency-egress decisions than on those of everyday decisions. This is obtained by considering the fact that the standard deviations of the coefficients for NEST1 and NEST2 variables in emergency model both are statistically significant at 99% level, while the standard deviation corresponding to NEST1 in the normal-egress model is insignificant. Exit choice models that do not consider this effect, may fail to accommodate the fact that some exits in the egress environment can be located closer to each other [10,39]. In other words, those models fail to predict the way exits are located in the egress geometry. The implication of this modelling shortcoming can be clarified by a simple example. Let us consider the choice scenario illustrated in Fig. 2(b). Let us also assume that, for some reason, Exit A that was previously chosen by the decision maker becomes extremely congested or blocked in a way that the individual excludes that exit from their choice set. As a result of this, the

Table 1
Modelling results.

Variable	Normal Egress Model				Emergency Egress Model			
	Estimated Parameter	Standard Error	*t*-Statistic	95% Confidence Interval	Estimated Parameter	Standard Error	t-Statistic	95% Confidence Interval
	Mean of Utility Coefficients				Mean of Utility Coefficients			
DIST	−0.037***	0.0026	−14.53	−0.042, −0.032	−0.049***	0.0031	−15.95	−0.055, −0.043
CROWD	−0.013***	0.0195	−6.48	−0.164, −0.088	−0.201***	0.019	−10.34	−0.239, −0.163
DIRFLOW1	−0.012	0.0296	−0.42	−0.070, 0.045	−0.067*	0.0368	−1.82	−0.139, 0.005
DIRFLOW2	−0.009	0.0225	−0.41	−0.053, 0.035	−0.027	0.0299	−0.89	−0.085, 0.032
VIS	0.688***	0.2188	3.14	0.259, 1.116	0.365**	0.1569	2.32	0.057, 0.672
DEST	2.966***	0.5373	5.52	1.913, 4.019	0.0 (Fixed Parameter)			
NEST1	0.0 (Fixed Parameter)				0.0 (Fixed Parameter)			
NEST2	0.0 (Fixed Parameter)				0.0 (Fixed Parameter)			
	Standard Deviations of Utility Coefficients				**Standard Deviations of Utility Coefficients**			
DIST	0.016***	0.0019	8.50	0.013, 0.020	0.026***	0.0020	12.78	0.022, 0.030
CROWD	0.145***	0.0154	9.38	0.114, 0.175	0.224***	0.0188	11.91	0.1868, 0.2604
DIRFLOW1	0.034	0.0442	0.77	−0.052, 0.120	0.2624***	0.0297	8.83	0.204, 0.321
DIRFLOW2	0.034	0.0442	0.77	−0.052, 0.120	0.2624***	0.0297	8.83	0.204, 0.321
VIS	1.519***	0.2294	6.62	1.070, 1.970	0.836***	0.1860	4.49	0.471, 1.200
DEST	2.453***	0.7385	3.32	1.006, 3.901	0.0 (Fixed Parameter)			
NEST1	0.668	0.5721	1.17	−0.453, 1.790	2.084***	0.1714	12.16	1.749, 2.420
NEST2	2.334***	0.2498	9.34	1.845, 2.824	1.739***	0.1641	10.59	1.417, 2.061
	Statistical Fit Measures				**Statistical Fit Measures**			
Number of Observations	1410				1410			
Initial Log-Likelihood	−2526.38				−2526.38			
Final Log-Likelihood	−1633.07				−1656.60			
McFadden ρ^2	0.35				0.34			

Note:
*** indicate statistical significance at 99% level respectively.
** indicate statistical significance at 95% level respectively.
* indicate statistical significance at 90% level respectively.

probability of other exits to be chosen will increase. However, it does not stand to reason to assume that the choice probability of all exits will increase by the same percent as the individual is more likely to change their decision to the nearest next available exit than to those located far away. Accordingly, the adjacent exit (i.e. Exit B, in this case) is more likely to receive a greater increase from the additional “market share” compared to other exits. Discrete choice models to which the relative positioning of the exits are not reflected through specification of explanatory variables or specification of the error terms will fail to properly capture this effect (for more mathematical details we refer to the relevant references [49]).

4.3.3. Distance versus crowding

The data suggests that the relative value that passengers place on choosing the closest exit (i.e. DIST variable) is by and large the same for normal and emergency situations. However, they place a far more negative value on the level of crowding around exit points (CROWD) in emergency evacuations compared to those of normal egress. This is also in line with this intuitive assumption that during emergencies, passengers are more likely to take longer routes if those lead them to less-crowded exit points. In other words, this quantifies the different levels of impatience that passengers may exhibit towards waiting for the congestions formed around exit areas to get cleared during normal and emergency situations. The data suggests a far higher level of impatience when egress is taking place in an emergency evacuation compared to an everyday egress scenario.

4.3.4. Herding and visibility

The normal-egress model suggests that in everyday wayfinding of passengers, directional flow (DIRFLOW) does not play a meaningful role. The mean and standard deviations for both DIRFLOW1 and DIRFLOW2 coefficients appear to be statistically insignificant (i.e. the hypothesis that mean and standard deviations of DIRFLOW coefficients are equal to zero cannot be rejected). This indicates that flows of passengers, whether towards visible exits or invisible exits, are typically neglected by other passengers for making exit decisions in everyday situations (i.e. they neither make their wayfinding decisions based on following other passengers nor do they tend to avoid them).

Referring to the estimation results of the emergency-egress model, however, one might draw the conclusion that the above proposition is true for emergency evacuations too. It should, however, be noted that the standard deviations for both factors (DIRFLOW1 and DIRFLOW2) are statistically significant even at 99% level. This means that directional flow does indeed have a meaningful effect on emergency wayfinding of passengers. However, the value they place on this factor is highly diverse, and those weights are distributed around zero in a nearly-balanced way that makes the central tendency of the estimates for those variables be close to zero.

According to the estimated utility coefficients for the emergency model (−0.067 and −0.027) majority of passengers perceive flows of pedestrians as a negative utility factor (i.e. as potential congestion or disutility factors). Moreover, they typically value flows of pedestrians moving towards visible and invisible exits differently. Speaking of the mean of coefficients, the coefficient they associate with the flows headed towards visible exits are nearly 2.5 times more negative than that of the flows headed to an invisible exit. In other words, a typical passenger is 60% likely to have a negative utility coefficient for DIRFLOW1 (i.e. a flow heading towards an exit point that the passenger can see); however the chance is 54% for DIRFLOW2 (i.e. a flow of the same size towards an exit area that the passenger cannot see). This suggests that the majority of people do not tend to exhibit herding behaviour during emergency evacuations; rather, they mostly tend to avoid joining or following large directional flows. However, the potential likelihood of herding to occur is higher when the passenger perceives a major flow rushing towards an exit which is invisible to him/her. This makes intuitive sense, from the standpoint that when the level of crowding around some exits are uncertain to decision makers, they are more likely to trust and follow other evacuees' decisions who can see those exits.

5. Discussion

In this section, we further discuss some aspects of this study in relation to the reliability and applicability of the data and models presented in this work.

5.1. Why hypothetical choices?

Question may arise with regard to the reliability of the data collected in hypothetical scenarios. In other words, scepticism might be expressed about the extent to which the data collected in hypothetical experiments could be trusted for prediction purposes. However, there are few points that need to be taken into account in relation with the provision of real choice data for the particular problem of crowd exit choice. One major challenge for collecting exit choice data based on observed choices of pedestrians made in real life is identification of the exact moments they make (or update) their decision as they exit a place. From a modelling perspective, this would be crucial to have access to the attribute levels of all exits at the moment they made their decision which makes the problem even more challenging. Moreover, such data might be extremely scarce for the emergency situations. Also, another advantage that the stated choice approach will offer for this particular problem is the possibility to stretch the attribute levels (such as level of crowding around exits) to extreme levels. Those extreme levels of attributes are something that might not be observable on a daily basis in public facilities, though might occur in exceptional cases. In order for the models to be usable for prediction purposes for those extreme scenarios, it is crucial that the source data contain information at those extreme levels. These basically constitute the main reasons the stated choice approach has been employed as a practical approach to capture the problem of crowd exit choice. The design of the stated-choice experiments with direct reference to participants' latest observed exit choices was indeed an effort to partly address the aforesaid shortcoming of hypothetical-scenario data. This, however, will by no means overshadow the importance of conducting a systematic validation process for the models constructed based on stated choices before using them for practical prediction purposes.

Also, it should be noted that our findings are based on the observations obtained from the responses of only 110 individuals in Melbourne, Australia. It would be interesting to see whether these behavioural findings can be replicated by independent samples particularly with participants of different cultural backgrounds than that of this study.

5.2. The effect of panic and irrationality in exit decisions

Another issue that can be argued about the data used in this study is the fact that decisions made by participants in the emergency exit choice data were made in a calm situation. This, of course, will restrict applicability of our proposed emergency model to the situations where the evacuation does not take place at extreme emergency levels which cause panic. The authors at this stage are not able to comment on the level of emergency that will cause irrationality in evacuees' decision as a result of panic and the way this may affect their behaviour. Clearly, the underlying features of escape under panic conditions are yet to be understood due to the absolute scarcity of the data in this context. Due to the ethical obstacles for conducting realistic experiments under panic, this aspect might still remain as a challenging topic in the literature. However, in the absence of any other data and information for exit choice under panic, the model presented in this work could be considered as an approximate modelling tool. The fact needs to be taken into consideration that even in panic escapes, the purpose of each individual evacuee still remain exiting as quickly as possible (similar to "non-panic" scenarios) although the level of optimality of the decisions made to achieve this end might be affected by the level of mental stress undergone during the evacuation.

5.3. The source and location of danger

The emergency exit choice model presented in this work did not include any variable indicating the location and source of danger as the causes of emergency. This is mainly due to the fact that inclusion of more explanatory variables would have necessitated each participant to answer to many more number of scenarios. This of course would have caused dramatic reduction in the participation rate and increased inattention to the scenarios as a result of fatigue. At least, in case the emergency evacuation is the result of fire, one practical solution to the modeller is to exclude form the simulation procedure the exits that require passing through the fire location from the virtual evacuees' choice set.

5.4. Why normal egress model is needed?

One can also raise the question about the necessity of developing the normal (i.e. nonemergency) exit choice model. The importance essentially arises from the fact that special events that attract unprecedented level of passenger demand to public transport places often require planners to conduct virtual simulation on the performance of their site. These are the cases where extreme levels of congestion make planners concerned about the efficiency of their crowd management policies and requires virtual simulations irrespective of the potential occurrence of emergency. From the planning perspective for special events, it is crucial to test multiple scenarios during the modelling to ensure that the space is tested at the extremes of its demand limits potential problems (e.g. blockage, congestion) are identified beforehand. Distinguishing the fundamental dissimilarities of pedestrian behaviour between normal and emergency cases provides the possibility of forecasting crowd egress under either scenario in a more accurate way. Currently, this modelling flexibility is not fully accommodated by the existing commercial crowd modelling packages.

6. Summary and conclusions

This study aimed to enhance the understanding of passengers' wayfinding behaviour in crowded public transport stations. This particularly concerns stations with complex architectural design that offer multiple exit points and routes to passengers. Significant progress has been made within the recent years in developing mathematical models for simulating escape behaviour of pedestrian crowd. Still, there is a very limited knowledge about the actual behaviour of people under different egress conditions based upon empirical data. This is a challenging problem as it is a matter in which human behaviour and decision making is involved. The diversity of the behaviour that different individuals may exhibit during egress situations makes the problem even more difficult to capture. This study attempted to contribute to bridging the gap that is believed to exist between developing mathematical modelling tools of crowd simulation (based on developers' intuitions or modeller-defined assumptions) and estimating the calibration parameters involved in those models based on empirical data [48]. It is believed that theoretically-simple crowd evacuation models whose parameters are calibrated using empirical data may be more likely to provide better predictions than sophisticated models whose governing rules are established based on a purely theoretical basis. This highlights the importance of provision of explanatory data in this context.

This work approached this problem by providing behavioural data using the state-of-the-practice methods of choice data collection recently proposed in econometrics literature. We designed surveys of stated-choice experiments, as a widely-practised data provision method that has served for several years as a pragmatic approach for analysing human choices [26,47], and sampled from passengers in a major public transport station. The hypothetical choice scenarios refer to the recent wayfinding experience of the participants. It is believed that this method makes respondents relate more realistically to the experiments and provide more reliable information, in contrast with the traditional approach of the stated-choice data collection in which experiments are designed on a purely hypothetical basis. The purpose of the survey was to obtain quantitative measures as to the way passengers make their exit decision and to estimate the weights (relative importance) they associate to different factors that can affect their decision. We examined the differences between the way individual passengers value those factors in normal (everyday) egress situations and emergency evacuations. Random-coefficient utility models were developed for both normal and emergency navigation of passengers. The mixed multinomial logit models developed in this study take into account the variability (heterogeneity) of people's preferences by allowing the utility coefficients to vary from individual to individual.

The key findings of our data analyses suggest that (1) there is combination of various factors contributing to the exit decisions that individuals make when exiting a built crowded environment and single factors such as choosing closest exits, or following other pedestrians cannot per se explain those decisions adequately. (2) Those contributing factors and their relative weights on pedestrian decisions are significantly different between normal and emergency choices of exit. (3) The utility weights individual passengers attach to a majority of contributing factors are significantly heterogeneous for both normal and emergency cases. (4) Proximity to destination is the most-dominant factor in everyday exit decisions (although not the sole determinant) whereas it proves to be completely irrelevant to the emergency decisions. (5) Passengers associate a much higher relative importance to avoiding heavily-crowded exit points during emergencies compared to normal exit situations. (6) Directional flows of passengers appear to not have a meaningful effect on everyday exit decisions neither in terms of mean nor standard deviation of the associated parameters. (7) During emergency evacuations, the presence of directional flows impact significantly on the decisions but in a highly heterogeneous way. Results indicate that narrow majority of passengers perceive negative utility (disutility) from directional flows and do not tend to follow flows of other pedestrians during emergency egress. (8) Moreover, directional flows proved to be perceived and evaluated by evacuees differently based on the visibility of the exit areas to which the flow is headed. Based on the modelling results, flows that are headed towards an exit that is invisible to the decision maker have more chance to be followed compared to the flows (of the same size) moving to an exit area that the decision maker can themselves see.

The proposed static models can be incorporated with a broad range of crowd simulation methods as alternatives for heuristic modeller-defined exit choice rules. They accommodate the dissimilarities of egress behaviour between emergencies and non-emergencies, thereby enabling planners to conduct virtual simulations under either scenario. The authors hope to report, in future works, on application of these models for forecast purposes (by integrating them with dynamic methods of crowd walking behaviour) in large stations and the way they impact on the accuracy and reliability of the forecasts compared to existing methods.

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